

ORIGIN STORY

Rewind a brain disease to the single cell where it began.

Someone will be diagnosed with Alzheimer's disease in 2046. The disease has already started.

It started in a cell that is alive right now, in a brain that feels completely normal, in a person who has no idea anything is wrong. By the time a doctor finally says the word, the process will have been running for twenty years — and the cell that started it will have been dead for most of them.

That twenty-year head start is the single most important fact in neurodegeneration, and it cuts both ways. It is why the field keeps arriving too late. It is also the most valuable unclaimed real estate in medicine: two decades of advance warning, if anyone could learn to read it.

Nobody can read it yet. Not Regeneron, not any pharma company, not any lab. That's why we're handing it to you.

You are a detective arriving at a crime scene twenty years late. The body is gone. All you have is the trail it left behind. Can you follow it backward?

Why this matters more than the average hackathon prompt

The scale is hard to overstate. The World Health Organization estimates 57 million people were living with dementia in 2021, with nearly 10 million new cases every year. Alzheimer's accounts for 60–70% of them. Dementia is the seventh leading cause of death worldwide, and in 2019 it cost the global economy around US\$1.3 trillion — roughly half of that borne by family members providing unpaid care ([WHO dementia fact sheet](#)).

For decades there was nothing. Now there is something — and the something is exactly the argument for this project.

The first drugs that actually modify the disease rather than mask symptoms have arrived. Lecanemab slowed cognitive and functional decline by 27% over 18 months in the CLARITY-AD trial. Donanemab showed roughly 35% slowing in TRAILBLAZER-ALZ 2. Both are real, both are FDA-approved, and both are modest.

Why only modest? Because they are given after diagnosis — after twenty years of accumulated damage. And the evidence increasingly points the same direction: **earlier is better**. In the 48-month extension of CLARITY-AD, the highest rates of stability and improvement were in patients who had the least tau pathology at baseline. The next generation of trials has drawn the obvious conclusion — TRAILBLAZER-ALZ 3 is testing donanemab in people who are cognitively normal but amyloid-positive. The whole field is racing backward in time toward the beginning.

But racing earlier has a floor, and the floor is this: you cannot treat what you cannot find. There is no method — none — for pointing at where in a brain the process actually began. That is the wall the field is currently standing at.

To be straight with you: you will not cure Alzheimer's this weekend. What you could plausibly do is build the first working version of a tool that asks the right question, using data anyone can download and math that fits on one page. That's a real contribution, and hackathons are unreasonably good at producing exactly that kind of thing — because the core method here is about thirty lines of linear algebra that nobody has seriously aimed at this problem yet.

The idea you're testing, in plain language

Alzheimer's and Parkinson's don't switch on everywhere at once. They start somewhere, and then they spread — cell to cell, along the brain's wiring, almost like an infection passing between neighbors.

The spreading part is well established. Pathologists have traced the route so reliably across thousands of autopsies that they gave the stages numbers, called Braak stages. And critically, the route follows neural connections rather than physical proximity: pathology jumps to a region that's wired to the source even if it's on the far side of the brain, and skips regions that are physically adjacent but not connected.

The part nobody has nailed down is the beginning. The leading hypothesis — the one this challenge is built around — is a two-stage story: **initiation**, where a single cell acquires a somatic mutation (a mutation it wasn't born with, that arose during your lifetime) and dies; followed by **propagation**, where misfolded protein templates itself onto neighboring connected cells and the cascade rolls outward.

Two halves. One is textbook. The other is a hole in the map.

Here's the part where it becomes a computer science problem

Strip the biology away and read the problem again: infer the origin of a spreading process on a network, given a snapshot taken long after it started.

You have almost certainly seen this before under a different name:

- Epidemic source detection — find patient zero in a contact network
- Rumor source localization — which node started the cascade?
- Inverse diffusion, or deconvolution on a graph

It is the same mathematics. The brain is the graph. Brain regions are the nodes. White-matter tracts are the edges. Pathology is the thing diffusing. If you have taken an algorithms or linear algebra course, you already have most of what you need.

And here is the gift: the standard human brain parcellation used in this kit has only 90 regions. That makes the inverse problem small enough to solve by exhaustive search — no gradient descent, no neural network, no training loop, no GPU. You can test every possible origin against every possible elapsed time in well under a second. Inverse problems are usually hard because the search space explodes. Yours doesn't.

The forward model: one equation

Let A be the connectivity matrix, D the diagonal matrix of row sums, and $L = D - A$ the graph Laplacian. (If you haven't met a Laplacian: it's just the matrix that says "each node leaks to its neighbors," and it's the discrete version of the heat equation. Heat spreading through metal, rumors through a network, pathology through a brain — same object.)

$$\frac{dx}{dt} = -\beta L x \quad \Rightarrow \quad x(t) = \exp(-\beta L t) \cdot x(0)$$

Set $x(0)$ to a single region — a one-hot vector, your candidate origin — pick an elapsed time t , and out comes a predicted pattern of pathology across all 90 regions. That's the network diffusion model from Raj et al. 2012, and it really is about five lines of NumPy.

The inverse model: the actual challenge

Now run it backward. You observe a pattern y — say, the regions affected at Braak stage IV. Which origin, and how long ago, best explains what you see?

$$\text{argmin over (seed } s, \text{ time } t) \quad \| \exp(-\beta L t) \cdot e_s - y \|^2$$

90 candidate seeds $\times$ 200 candidate timepoints = 18,000 evaluations. Precompute a single eigendecomposition and each evaluation collapses to one matrix–vector product. So you don't get one answer — you get a ranked, scored list of every candidate origin in the brain, which happens to visualize as a gorgeous heat map.

Ninety minutes to something real

Do this before you get ambitious. It puts a genuine result on genuine human data in your hands, and every track below builds on it.

Step 1 — Get a real human connectome (no account, no approval)

Download the 88-subject human structural connectivity dataset from OSF: <https://osf.io/yw5vf/> — Škoch et al., Scientific Data 2022, released CC BY 4.0. No login, no data use agreement, no waiting. Grab the `structural_connectivity_matrices/` folder (one 90 $\times$ 90 matrix per subject) and `AAL_regions.csv`, which lists the region names in matrix order. The dataset paper is open access at [nature.com/articles/s41597-022-01596-9](https://www.nature.com/articles/s41597-022-01596-9).

Step 2 — Run this

```
import numpy as np, pandas as pd
from scipy.linalg import eigh

# NOTE: files are named .csv but are TAB-delimited. This eats teams alive.
A = pd.read_csv('S001.csv', sep='\t', header=None).values
A = (A + A.T) / 2 # symmetrize (see Landmines)
regions = pd.read_csv('AAL_regions.csv', header=None)[0].tolist()

L = np.diag(A.sum(1)) - A # graph Laplacian
lam, U = eigh(L) # symmetric -> real eigendecomposition

def spread(seed, t, beta=1.0):
    """Pattern at time t from a point seed: exp(-beta*L*t) @ e_seed"""
    return U @ (np.exp(-beta * lam * t) * U[seed, :])
```

```
# FORWARD: watch pathology leave the hippocampus
s = regions.index('Hippocampus_L')
for t in [0.1, 0.5, 1.0, 2.0]:
    top = np.argsort(spread(s, t))[:-1][:5]
    print(t, [regions[i] for i in top])

# INVERSE: 90 seeds x 200 times. Brute force. Sub-second.
observed = ... # your target pattern, length 90
cands = sorted((np.linalg.norm(spread(s, t) - observed), regions[s], t)
               for s in range(90)
               for t in np.linspace(0.01, 5, 200))
print('most likely origin:', cands[0])
```

Step 3 — Validate against something already known

This is the step that wins you the room. Braak staging already tells us roughly where Alzheimer's tau pathology begins — the entorhinal and parahippocampal region — before it moves to hippocampus, then limbic areas, then neocortex. So hand your inverse model a late-stage pattern and see whether it points home.

If it does, you can demonstrate in ten seconds that your method recovered an answer that took neuropathologists decades to establish. If it doesn't, that's a finding too — say so plainly and explain what you think went wrong. Both versions are respectable. Only pretending is not.

Step 4 — Make it visual

One line of nilearn gives you a self-contained interactive 3D brain as an HTML file you can drop into a web page. See the toolkit section below.

What's established, what's precedent, what's wide open

Be precise about this in your pitch. Knowing which half of the hypothesis is textbook and which half is a hole is most of what separates a strong project from a hand-wave.

Claim	Confidence	Why
Misfolded proteins spread cell to cell along neural connections	Established	Tau, α -synuclein, TDP-43 and SOD1 all show templated misfolding and cell-to-cell transmission in cell culture, organoids and animal models. Inject seeds into one region and pathology appears in connected regions — following wiring, not physical distance.
The spread route is stereotyped and stageable	Established	This is the basis of Braak staging in Alzheimer's and Parkinson's: a reproducible anatomical ordering observed across thousands of autopsies.
A somatic mutation in a single cell can initiate neurodegeneration	Strong precedent	Somatic CAG expansion in HTT: the repeat grows from ~40 to 100–500+ copies in individual striatal projection neurons over decades, and past roughly 150 repeats the cell loses its identity and dies (Handsaker et al., Cell 2025). Somatic mtDNA deletions also accumulate in neurons with age and are enriched in Parkinson's substantia nigra.

Claim	Confidence	Why
The initiating event can be inferred from the propagation pattern	OPEN — this is your target	Nobody has shown this. There is no published answer key to check yourself against, which is unusual for a hackathon problem and is precisely the point.

Why finding the beginning is so much harder than tracing the spread

Huntington's disease is the one case where we caught the initiating event, and we caught it because of a lucky accident of biology: the CAG repeat takes decades to expand, and it has to cross a threshold before the cell dies. That slow fuse leaves a long window in which you can catch cells mid-process — some at 60 repeats, some at 200 — and reconstruct the sequence from a single snapshot.

Most candidate initiating mutations probably don't come with a slow fuse. A cell acquires the mutation and dies. No window, no gradient, no evidence left behind. By the time the disease is detectable, the cells that started it are gone — which is the whole reason this is a detective problem rather than a measurement problem.

There are two honest ways forward, and both are fair game: **catch the event directly** — find a measurement or a study design that could detect the first cell before it dies — or **use the propagation to predict the initiation** — work backward from what's left. The second is more tractable in a weekend. The first is more original. Nobody will be upset if you attempt the first and fail interestingly.

Real data you can have in the next ten minutes

Read this before you choose a dataset. The most common way a biomedical hackathon project dies is waiting on data access that was never going to arrive in time. Everything in Tier 1 is downloadable right now with no account and no approval. Start there. Treat Tier 3 as reading material, not as a plan.

Tier 1 — open, immediate, no account

Start with one of these three

Human structural connectome — 88 subjects, 90×90 AAL matrices, CC BY 4.0. The recommended entry point, and what the code above expects. <https://osf.io/yw5vf/>

Allen Mouse Brain Connectivity Atlas — a directed connectome. Mesoscale projection strengths between mouse brain structures, built from viral tracer injections. Unlike diffusion MRI this one has genuine directionality, which matters a lot if you care which way pathology flows. Portal:

<https://connectivity.brain-map.org> · Python: `pip install allensdk`, docs at

<https://allensdk.readthedocs.io> · Worked notebook:

alleninstitute.github.io/AllenSDK/static/examples/nb/mouse_connectivity.html

SEA-AD (Seattle Alzheimer's Disease Brain Cell Atlas) — real human disease tissue. Single-nucleus RNA-seq and ATAC-seq, spatial transcriptomics, and quantified neuropathology across donors spanning the whole spectrum from neurotypical to severe Alzheimer's. Genuinely open. This is the best human AD resource obtainable without paperwork, and the only one here that touches actual diseased human

brain. Portal: <https://portal.brain-map.org/explore/seattle-alzheimers-disease> · Docs and downloads: <https://brain-map.org/consortia/sea-ad/our-data> · AWS Open Data buckets: <https://registry.opendata.aws/allen-sea-ad-atlas/> · Browse interactively on CELLxGENE: cellxgene.cziscience.com/collections/1ca90a2d-2943-483d-b678-b809bf464c30

More open connectomes

- **HCP functional connectivity matrices** — group-level network matrices from the Human Connectome Project, fetched by a single nilearn function call, no registration:
`nilearn.datasets.fetch_megatrawls_netmats()`. Example:
nilearn.github.io/stable/auto_examples/01_plotting/plot_visualize_megatrawls_netmats.html
- **C. elegans connectome** — the complete wiring diagram of a 302-neuron nervous system (the commonly analysed somatic subgraph is about 280 connected neurons). Small enough to simulate exactly, cell by cell. OpenWorm's Connectome Toolbox packages every published version with Python readers: <https://openworm.org/ConnectomeToolbox/> · Raw adjacency matrices from the Emmons lab: <https://wormwiring.org/pages/adjacency.html>
- **FlyWire — whole adult fly brain connectome** — ~130k neurons with synaptic resolution, public static snapshots, bulk downloadable: <https://codex.flywire.ai> (see the [FAQ](#) for which release to use) · Cell type annotations: https://github.com/flyconnectome/flywire_annotations
- **netneurotools** — parcellations, consensus networks and network-neuroscience utilities: <https://netneurotools.readthedocs.io>

Genes, variants, expression

- **abagen** — maps Allen Human Brain Atlas gene expression onto whatever parcellation you're using, so you can ask "what's biologically distinctive about the region my model picked?" `pip install abagen` · <https://abagen.readthedocs.io> · Underlying atlas: <https://human.brain-map.org>
- **Open Targets Platform** — gene–disease association scores with evidence, no API key. "Which genes are linked to Alzheimer's, and how strongly?" is one query. Endpoint: <https://api.platform.opentargets.org/api/v4/graphql> · Interactive playground: api.platform.opentargets.org/api/v4/graphql/browser · Docs: platform-docs.opentargets.org/data-access/graphql-api
- **GTEx Portal** — bulk gene expression across 13 distinct human brain regions, open REST API, no key. Note this is expression data, *not* a somatic mutation catalog. API docs: <https://gtexportal.org/api/v2/redoc> · Overview: <https://gtexportal.org/home/apiPage> (v1 is discontinued — use v2)
- **CELLxGENE Discover** — hundreds of standardised single-cell datasets including human brain, browsable and downloadable: <https://cellxgene.cziscience.com>
- **Variant and gene annotation, all open and key-free** — Ensembl REST (<https://rest.ensembl.org>), MyVariant.info (<https://myvariant.info>), MyGene.info (<https://mygene.info>), NCBI E-utilities (<https://www.ncbi.nlm.nih.gov/books/NBK25501/>)

Literature and trials — for the agentic and NLP tracks

- **Europe PMC REST API** — full-text search across the biomedical literature, open: <https://europepmc.org/RestfulWebService>

- **ClinicalTrials.gov API v2** — every registered trial, including every Alzheimer's trial that ever failed: <https://clinicaltrials.gov/data-api/api>
- **OpenNeuro** — thousands of BIDS-formatted neuroimaging datasets, most fully open: <https://openneuro.org>. Use if you want raw imaging rather than precomputed matrices, but budget time for preprocessing.

Tier 2 — free account, about five minutes

- **neuPrint (Janelia)** — hemibrain, MANC and maleCNS connectomes with a real query API and cell-type resolution. Register, copy a token, `pip install neuprint-python`: <https://neuprint.janelia.org>

Tier 3 — do not start here

These are excellent datasets and the right answer for a PhD project. They are the wrong answer for a weekend, and we'd rather tell you now than have you discover it on Saturday night.

- ADNI (Alzheimer's Disease Neuroimaging Initiative) — requires an application, an institutional affiliation, and a signed data use agreement. Review takes roughly two weeks. HackMIT is in two weeks.
- PPMI (Parkinson's Progression Markers Initiative) — same shape, same timeline.
- NIMH Data Archive (including Brain Somatic Mosaicism Network data), NIAGADS, and the controlled-access tiers of NeMO — all require approved data use requests countersigned by an institutional official.
- UK Biobank — months, and it costs money.

If you specifically want human somatic mutation data: this is the honest gap in the whole challenge. Human brain somatic mosaicism data sits almost entirely behind controlled access, because single-cell genotypes are identifying by nature. You will not get it this weekend. That's a real constraint rather than a failure on your part — and a project that names the constraint and routes around it intelligently is stronger than one that pretends it isn't there.

Five ways in

Pick one. Please don't pick three. With over a thousand hackers in the room, what distinguishes a project is depth and a demo that survives contact with a judge — not breadth.

1 — Inverse Connectome: the brain map that points at the origin

The flagship build, and what the ninety-minute path sets you up for. A user clicks a brain region or selects a disease stage; your app runs the inverse diffusion model and lights up the predicted origin with a confidence ranking across all 90 regions. Then it animates the inferred spread. The money shot is playing that animation in reverse, collapsing everything back into one glowing region.

Stretch: handle multiple simultaneous seeds; add a production term so pathology grows rather than merely redistributing; run the same inference on the human diffusion-MRI connectome and the directed mouse tracer connectome and explain why they disagree.

2 — Pseudotime: find the first domino in real human tissue

Skip the connectome entirely. SEA-AD gives you human donors spanning the full continuum from neurotypical to severe Alzheimer's, with pathology quantified per brain region and single-cell data alongside. Order those donors by pathology burden to construct a pseudo-timeline, then ask which region — or which cell type — starts changing first. Initiation detection from real human data, with no assumptions baked in about how spread works.

This is the only track that touches actual diseased human brain. Real data beats a prettier simulation, every time.

3 — Single-cell resolution: a whole brain small enough to solve exactly

The region-level model is a blunt instrument. The hypothesis is about one cell, and "parahippocampal gyrus" contains millions of them. So change organism. *C. elegans* has a completely mapped nervous system of 302 neurons. You can simulate templated protein spread cell by cell, animate every single neuron, and then test whether your inverse method recovers the exact neuron you seeded — ground truth, because you created it.

Best ratio of demo quality to honesty of claim on this list: a real connectome, an exact simulation, a verifiable answer, and 302 glowing dots is a beautiful thing to watch. Scale to FlyWire if you have time left.

4 — Detectability: what measurement would actually catch it?

Invert the question. Don't predict where it started — work out what it would take to watch it happen. Given a plausible model of a single mutant cell dying and seeding its neighbors, simulate a screening program: how often would you need to sample, at what anatomical resolution, with what assay sensitivity, to catch the initiating event before the cascade becomes irreversible? Deliver a study design with the tradeoffs quantified.

Almost nobody will build this, and it is the question a drug company actually needs answered. High risk, highest ceiling.

5 — Language and agents on top of the model

Put a human-friendly surface on the science. Someone describes symptoms in plain words; you map that to a likely pathology stage, run the inverse model, and hand back an interpretable origin map with a readable explanation. Or point an agent at Europe PMC to mine reported regional pathology patterns out of thousands of papers and automatically assemble the observed-pattern vectors your model consumes — turning a manual data-entry bottleneck into a pipeline.

Warning: an LLM wrapper with nothing underneath is the single most common thing judges get shown all weekend. The language layer has to sit on top of real computation.

Toolkit: making it look like the future

A correct model that demos badly loses to a decent model that demos beautifully. Budget real time for this — but build it on top of something that genuinely computes.

- **nilearn — fastest credible brain render.** `plotting.view_img()` and `view_surf()` emit self-contained interactive 3D HTML. Roughly one line. Also ships atlases, surface meshes and dataset fetchers. `pip install nilearn` · <https://nilearn.github.io/stable/plotting/index.html>
- **Plotly — most reliable under time pressure.** A 3D scatter of AAL region centroids, sized and colored by predicted pathology. Half an hour, looks sharp, very hard to break on stage. <https://plotly.com/python/3d-scatter-plots/>
- **NiiVue — WebGL brain rendering in the browser.** Handles volumes, meshes, tractography and connectomes natively; over 30 formats. <https://github.com/niivue/niivue>
- **Cytoscape.js / sigma.js / D3 — the connectome as a graph.** For animating edges lighting up as pathology propagates. <https://js.cytoscape.org>
- **three.js — custom 3D.** Pair it with the fsaverage surface mesh from nilearn if someone on your team has done shader work. <https://threejs.org>
- **Streamlit or Gradio — app shell in under an hour.** Spend the time you save on the model. <https://streamlit.io> · <https://gradio.app>
- **Graph machinery.** NetworkX (<https://networkx.org>) for general graph work; `bctpy` (<https://github.com/aestrivex/bctpy>) for the standard brain-network measures; `scipy.linalg.eigh` and `expm` for the diffusion math.

The single best demo move: run the spread animation in reverse. Start from the late-stage pattern your model was handed, and play time backward until everything collapses into one region. It costs about twenty extra lines and it is the thing people will remember when they're deciding.

Landmines (each of these has cost someone four hours at 3am)

1. The connectivity files are named `.csv` but they are tab-delimited. Pass `sep='\t'`.
2. The AAL-90 parcellation has no entorhinal cortex label. The nearest proxies for the Braak stage I–II origin are `ParaHippocampal_L/R` and `Hippocampus_L/R`. State which proxy you used and why.
3. AAL-90 also has no brainstem and no substantia nigra. Parkinson's pathology is thought to begin in brainstem and olfactory structures, so the PD origin is not representable in this parcellation at all. Use Alzheimer's/tau for the AAL demo, or switch to the mouse atlas for PD.
4. This dataset deliberately excludes the cerebellum, so the matrices are 90×90 , not 116×116 .
5. The matrices are nearly but not exactly symmetric, and that asymmetry does NOT encode axonal direction — diffusion MRI is fundamentally undirected. Symmetrize with $(A + A.T)/2$. If you want genuine directionality, use the Allen mouse tracer data.
6. Tractography produces false-positive connections. Threshold weak edges, and report the threshold you chose.

7. The 88 subjects are healthy controls from a schizophrenia study. There are no Alzheimer's or Parkinson's patients in this dataset — you are using a healthy connectome as the substrate and Braak staging as the observed pattern. Put that on a slide. A judge who notices before you say it will discount everything after.
8. Laplacian diffusion conserves total mass. If your predicted pattern sums to the same value at every t , that's the math working correctly, not a bug. Add a source term if you want growth.
9. Don't call `scipy.linalg.expm` inside your search loop. Eigendecompose L once and reuse it — that's the difference between sub-second and several minutes.
10. Do not claim your tool diagnoses, predicts or stages disease in real patients. It doesn't, everyone in the room knows it doesn't, and claiming it costs more credibility than the feature could ever earn.

What we'll be impressed by

- It runs. Live. From a cold start, on a laptop that isn't the one you built it on.
- You validated against something already known — recovering the parahippocampal region as the origin of AD tau pathology is a ten-second, unarguable demonstration that your method does something real.
- You clearly labeled what is measured data and what is simulated. Every time. This buys enormous trust.
- You can name one thing your model gets wrong, and explain why.
- You made a deliberate decision about what NOT to build, and can say what it was.
- You found an angle we hadn't thought of. The five tracks above are suggestions, not a menu.

What won't land

- A beautiful animation with nothing computing behind it. We will ask what happens if we change the input.
- Twelve machine learning models, none of them validated against anything.
- Any claim of clinical utility.
- A polished deck and no running code.
- "We applied for dataset access and are waiting to hear back."

One piece of unsolicited advice

Stop adding features before you think you need to, and spend that time making the demo path bulletproof and rehearsing it out loud. The teams that fall short on Sunday morning are almost never the ones with the weakest idea. They're the ones who were still writing code when judging started.

Six things to read, ranked by usefulness at 2am

1. **Raj, Kuceyeski & Weiner (2012), "A network diffusion model of disease progression in dementia," *Neuron*.** This is your forward model, essentially verbatim. Read this one first.

2. **Škoch et al. (2022), Scientific Data 9:486.** The dataset paper for the connectome you're downloading. Read the Usage Notes section specifically — it will save you from three of the landmines above. [nature.com/articles/s41597-022-01596-9](https://www.nature.com/articles/s41597-022-01596-9)
3. **Handsaker et al. (2025), "Long somatic DNA-repeat expansion drives neurodegeneration in Huntington's disease," Cell.** Open access. The closest thing that exists to proof of the initiation half of the hypothesis, and a genuinely beautiful piece of work. [cell.com/cell/fulltext/S0092-8674\(24\)01379-5](https://www.cell.com/cell/fulltext/S0092-8674(24)01379-5)
4. **Braak & Braak (1991) for Alzheimer's; Braak et al. (2003) for Parkinson's.** You need the stage ordering. Skim for the anatomical sequence and don't get lost in the histology.
5. **Brockmann & Helbing (2013), "The hidden geometry of complex, network-driven contagion phenomena," Science.** Effective distance — an elegant alternative to the Laplacian approach for locating an origin on a network.
6. **Shah & Zaman (2011), "Rumors in a network: Who's the culprit?"** The pure CS framing of your problem, with provable guarantees. Useful if you'd rather have a principled estimator than a brute-force search.

Come talk to us

Regeneron mentors are accessible for the whole event (mostly via Discord channels and Google Meet, but Henry will be in-person periodically, and/or staying nearby in walking distance). Find us early rather than late — the most valuable thing we can do is tell you in five minutes that an idea won't work, before you spend six hours on it. We would much rather field a naive biology question than watch a good team quietly build the wrong thing!